



# AI Learning Buddy

## Assignment Brief & Submission Guidelines

### 1. Objective

Design a simple AI Learning Buddy for a topic of your choice.

### 2. Problem Statement

Learners often need help understanding a new topic. Your objective is to design an AI learning buddy that explains concepts, gives examples, asks quiz questions, and provides feedback.

### 3. Tools / Concepts to Use

- ChatGPT / Gemini / Copilot
- Prompt templates
- Role prompting
- Quiz generation
- Responsible AI checks
- Streamlit, for technical learners who want to turn this into a working web app

### 4. The Task

Your assignment, in one line, is:

*"Create an AI learning buddy for a topic of your choice. The AI buddy should explain the topic in simple language, provide one real-life example, ask quiz questions, and give feedback. Submit the prompt templates, sample outputs, and the web page built using Streamlit."*

## 5. Deliverables Checklist

Every item below must be completed and included in your submission. Work through them in order — each one builds on the last.

Topic selected

AI Buddy Persona description

5 Reusable Prompt Templates

One full sample learning conversation (screenshot or text)

5-question quiz + answers

Your reflection on AI limitations (300–400 words)

Streamlit app link (Mandatory)

## 6. What Each Deliverable Should Contain

### 6.1 Topic Selected

Pick one clear, specific topic — not too broad. "Photosynthesis" works better than "Biology." "Binary Search" works better than "Algorithms."

### 6.2 AI Buddy Persona Description

Write a short description (3–5 sentences) of the personality and role you are giving your AI buddy. This is where role prompting comes in — for example, defining it as a patient, encouraging tutor for beginners, or an exam-focused quiz master. Include the actual persona/system prompt you used.

### 6.3 Five Reusable Prompt Templates

Design 5 prompt templates that could work for any topic, not just the one you chose (swap out the topic name and they should still work). At minimum, cover these functions:

1. A prompt to explain the topic in simple language.
2. A prompt to give one real-life example.
3. A prompt to generate quiz questions.

4. A prompt to evaluate/give feedback on a learner's answer.
5. Full Session Prompt

## 6.4 Sample Learning Conversation

Submit one complete conversation showing your AI buddy in action from start to finish — the learner's questions/topic, and the AI's responses. A screenshot of the chat, or a clean copy-pasted transcript, both work.

## 6.5 Five-Question Quiz + Answers

Include the AI-generated quiz (5 questions, based on your chosen topic) along with the correct answers, generated using your quiz prompt template (Attach Screenshots)

## 6.6 Reflection on AI Limitations (300–400 words)

Write a short, thoughtful reflection covering:

- Strengths — what are the strengths of AI as a tutor?
- Limitations — what are the limitations of AI-generated explanations?
- Improvements — how can we improve AI learning tools?

### Reflection tips

- Use specific examples from your own sample conversation, not just general statements.
- Stay within the 300–400 word range — this is a reflection, not an essay.

## 6.7 Streamlit App Link (Mandatory)

Build a working web page version of your AI Learning Buddy using Streamlit (following the build guide provided separately), deploy it, and submit the public link. This deliverable is mandatory for all students/

# 7. Submission Instructions

6. Complete all seven deliverables listed in Section 5.
7. Compile them into a single document (or a shared folder containing all files/screenshots).
8. Submit everything through the submission link: <https://forms.gle/rLt4i4mmVpiyFQv79>
9. Double-check your Streamlit app link opens correctly in a fresh browser tab before submitting.

# 8. Evaluation Rubric

| Criterion      | Weight | What we're looking for                                              |
|----------------|--------|---------------------------------------------------------------------|
| Persona design | 15%    | Clear, well-defined AI buddy persona with an effective role prompt. |

|                     |     |                                                                                                      |
|---------------------|-----|------------------------------------------------------------------------------------------------------|
| Prompt templates    | 20% | 5 well-structured, genuinely reusable templates covering the required functions.                     |
| Sample conversation | 15% | Complete, coherent conversation demonstrating the buddy actually working.                            |
| Quiz quality        | 15% | 5 relevant questions with correct, clearly presented answers.                                        |
| Reflection          | 15% | Thoughtful, specific reflection covering strengths, limitations, and improvements.                   |
| Streamlit app       | 20% | App is live, functional, and matches the required features (topic input, activity choice, response). |

***Good luck!***